

This project analyzes supermarket transaction data (1,000 records, 17 variables) using Python (pandas, seaborn, matplotlib) to explore sales performance and customer patterns.

The dataset is clean, with no missing values, and contains details on product lines, customer type, gender, payment methods, pricing, and ratings.

Total sales amount to about 322,967 with an average sale of roughly 323 per transaction, indicating relatively high basket values in this supermarket sample.

Food and beverages generate the highest revenue, followed closely by sports and travel, electronic accessories, fashion accessories, and home and lifestyle, while health and beauty lags slightly behind.

Female customers contribute a higher share of total sales than male customers, suggesting a more valuable customer segment in revenue terms.

Sales are fairly balanced across payment methods, with cash slightly ahead of e-wallets and credit cards, indicating customers use a mix of traditional and digital payments.

Exploratory visualizations such as bar charts, boxplots, and a correlation heatmap help reveal which product lines and customer segments drive revenue and how numeric variables (like unit price, quantity, and rating) relate to overall sales and profitability.